In [23]:
```
if (!require("pacman")) install.packages("pacman")
pacman::p_load(knitr, tidyverse, ggplot2, dplyr, lubridate, stringr, readxl,
source("../data-build/functions.R")
source("../data-build/rating_variables.R")

ma_final <- read_csv("../../hwk3/data/output/final-ma-data.csv")
```

```
Rows: 440877 Columns: 110
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (16): contractid, state, county, org_type, plan_type, partd, snp, eghp,
...
dbl (94): planid, fips, n_nonmiss, avg_enrollment, sd_enrollment, min_enrol
l...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this mes
sage.
```

Question 1

In [20]:
```
summary_table <- ma_final %>%
    filter(!is.na(Star_Rating)) %>%
  mutate(plan_market_share = avg_enrollment / avg_eligibles) %>%
  group_by(year) %>%
  summarize(
    `Mean Star Rating` = mean(Star_Rating, na.rm = TRUE),
    `Mean Enrollment` = mean(avg_enrollment, na.rm = TRUE),
    `Mean Market Share` = mean(plan_market_share, na.rm = TRUE),
    `Plan Count` = n()
  )

summary_table
```

A tibble: 6 × 5

year	Mean Star Rating	Mean Enrollment	Mean Market Share	Plan Count
<dbl>	<dbl>	<dbl>	<dbl>	<int>
2010	2.969968	345.9702	0.01245348	60519
2011	3.255317	388.3079	0.01506869	55249
2012	3.359556	410.6486	0.01656644	58169
2013	3.548875	406.7334	0.01631573	64174
2014	3.806049	451.1140	0.01785064	58636
2015	3.961542	482.5712	0.01930427	60819

Question 2

In [22]:
```
# Filter for plans without a star rating
no_stars_plan_year <- ma_final %>%
  filter(is.na(Star_Rating)) %>%
  group_by(year, contractid, planid) %>%
  summarize(
    # Total enrollment for the plan across its service area
    total_plan_enrollment = sum(avg_enrollment, na.rm = TRUE),
    # Total MA enrollment in the counties served by this plan
    total_market_enrollment = sum(avg_enrolled, na.rm = TRUE),
    .groups = "drop"
  ) %>%
# Market Share for each plan
  mutate(plan_market_share = total_plan_enrollment / total_market_enrollment

# Create table for unrated plans
summary_no_stars <- no_stars_plan_year %>%
  group_by(year) %>%
  summarize(
    `Mean Enrollment` = mean(total_plan_enrollment, na.rm = TRUE),
    `Mean Market Share` = mean(plan_market_share, na.rm = TRUE),
    `Count of Plans` = n()
  )

summary_no_stars
```

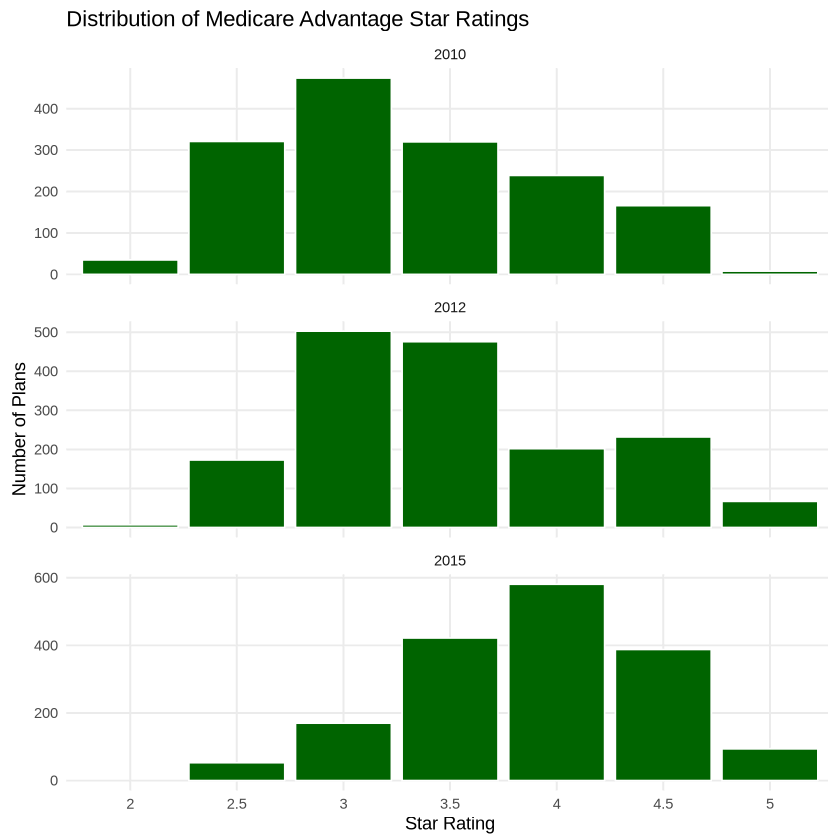
A tibble: 6 × 4

year	Mean Enrollment	Mean Market Share	Count of Plans
<dbl>	<dbl>	<dbl>	<int>
2010	1289.6567	0.009691769	964
2011	1276.4435	0.016652349	649
2012	1189.5571	0.012039689	494
2013	764.2311	0.014137061	409
2014	970.8587	0.015523844	489
2015	1121.8427	0.010754724	533

Question 3

In [34]:
```
star_dist_data <- ma_final %>%
  filter(year %in% c(2010, 2012, 2015)) %>%
  distinct(year, contractid, planid, Star_Rating) %>%
  filter(!is.na(Star_Rating))

ggplot(star_dist_data, aes(x = factor(Star_Rating))) +
```

```
  geom_bar(fill = "darkgreen", color = "white") +
  facet_wrap(~year, ncol = 1, scales = "free_y") +
  theme_minimal() +
  labs(
    title = "Distribution of Medicare Advantage Star Ratings",
    x = "Star Rating",
    y = "Number of Plans"
  ) +
    theme(panel.grid.minor = element_blank())
```



The distribution has shifted rightward over time, showing a general increase in plan ratings. In 2010, the majority of MA ratings were concentrated at 3. This held true in 2012 with an additional increase in 3.5 ratings. By 2015, we see that the highest number of plans are 4 stars (~600), with 3.5 and 4.5 stars close behind at ~400 plans each.

Question 4

In [5]:
```
reg_data <- ma_final %>%
  group_by(year, contractid, planid) %>%
  summarize(
    star_rating = mean(Star_Rating, na.rm = TRUE),
    plan_enrollment = sum(avg_enrollment, na.rm = TRUE),
    total_market_enrollment = sum(avg_enrolled, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(market_share = plan_enrollment / total_market_enrollment) %>%
  filter(!is.na(star_rating), !is.na(market_share)) %>%
```

```
  # Create star rating factor (Excluded Group = 2.5 or below)
  mutate(star_cat = case_when(
    star_rating <= 2.5 ~ "stars_2.5_or_below",
    star_rating == 3.0 ~ "stars_3.0",
    star_rating == 3.5 ~ "stars_3.5",
    star_rating == 4.0 ~ "stars_4.0",
    star_rating >= 4.5 ~ "stars_4.5_plus"
  )) %>%
  mutate(star_cat = factor(star_cat, levels = c("stars_2.5_or_below", "stars

# Run regression for each year
models_list <- map(2010:2015, ~ lm(market_share ~ star_cat, data = filter(re

# Table of results using stargazer
stargazer(models_list,
          type = "text",
          column.labels = as.character(2010:2015),
          dep.var.labels = "Market Share",
          covariate.labels = c("3 Stars", "3.5 Stars", "4 Stars", "4.5+ Star
          omit.stat = c("f", "ser"),
          title = "OLS Regression of Market Share on Star Ratings (2010-2015
          align = TRUE)
```

```
OLS Regression of Market Share on Star Ratings (2010-2015)
==================================================================
                              Dependent variable:
             ----------------------------------------------------
                                   Market Share
               2010      2011      2012      2013     2014      2015
               (1)       (2)       (3)       (4)      (5)       (6)
------------------------------------------------------------------
3 Stars       0.007*    0.010*    0.0001   0.016***  0.006    -0.005
             (0.004)   (0.005)   (0.005)   (0.005)  (0.011)  (0.007)

3.5 Stars    0.013***   0.009    -0.003   0.011**   0.006    0.002
             (0.004)   (0.005)   (0.005)   (0.005)  (0.011)  (0.007)

4 Stars      0.014***   0.006    -0.008    0.005    0.003    -0.006
             (0.004)   (0.006)   (0.005)   (0.005)  (0.011)  (0.007)

4.5+ Stars   0.019***  0.016***  -0.005    0.008   -0.002    -0.005
             (0.005)   (0.006)   (0.005)   (0.005)  (0.011)  (0.007)

Constant     0.013***  0.019***  0.030***  0.015***  0.019*   0.026***
             (0.003)   (0.005)   (0.004)   (0.004)  (0.010)  (0.007)

------------------------------------------------------------------
Observations  1,563     1,511     1,660     1,863    1,767    1,708
R2            0.015     0.005     0.003     0.008    0.005    0.004
Adjusted R2   0.013     0.002     0.0002    0.006    0.002    0.002
==================================================================
Note:                                  *p<0.1; **p<0.05; ***p<0.01
```

Question 5

In [18]:
```
# Filter for 2010
ma_2010 <- ma_final %>%
  filter(year == 2010) %>%
  select(contractid, planid, Star_Rating, avg_enrollment, plan_type, partd,

# Calculate running variable (raw score) using rowmeans
rd_table_data <- ma_2010 %>%
  mutate(
    raw_score = rowMeans(
      select(., any_of(rating.vars.2015)) %>% select(where(is.numeric)),
      na.rm = TRUE
    )
  ) %>%
  filter(!is.na(raw_score)) %>%
  mutate(
    raw_decimal = raw_score - floor(raw_score),
    # CMS Rounding Rules
    rounded_up = case_when(
      raw_decimal >= 0.25 & raw_decimal < 0.50 & Star_Rating > floor(raw_sco
      raw_decimal >= 0.75 & Star_Rating > (floor(raw_score) + 0.5) ~ TRUE,
      TRUE ~ FALSE
    )
  )

# Create summary table
rounding_summary <- rd_table_data %>%
  filter(rounded_up == TRUE) %>%
  group_by(Star_Rating) %>%
  summarize(`Number of Plans Rounded Up` = n()) %>%
  rename(`Star Rating` = Star_Rating) %>%
  arrange(`Star Rating`)

rounding_summary
```

A tibble: 7 × 2

Star Rating	Number of Plans Rounded Up
<dbl>	<int>
2.0	243
2.5	10601
3.0	3970
3.5	4122
4.0	2376
4.5	1310
5.0	97

Question 6

In [19]:
```
# Calculate raw scores
rd_table_data <- ma_2010 %>%
  mutate(
    raw_score = rowMeans(
      select(., any_of(rating.vars.2015)) %>% select(where(is.numeric)),
      na.rm = TRUE
    )
  ) %>%
  filter(!is.na(raw_score))

# Aggregate to plan-year level
rd_enrollment_data <- rd_table_data %>%
  group_by(contractid, planid) %>%
  summarize(
    raw_score = mean(raw_score, na.rm = TRUE),
    total_enrollment = sum(avg_enrollment, na.rm = TRUE),
    .groups = "drop"
  )

# RD estimation function (bandwidth = 0.125)
estimate_rd <- function(data, threshold, bw = 0.125) {
  data %>%
    mutate(dist = raw_score - threshold) %>%
    filter(abs(dist) <= bw) %>%
    mutate(treated = ifelse(dist >= 0, 1, 0)) %>%
    lm(total_enrollment ~ treated + dist, data = .)
}

# Run regressions for the specific thresholds (2.75 is the cutoff for 3.0 st
model_30 <- estimate_rd(rd_enrollment_data, 2.75)
model_35 <- estimate_rd(rd_enrollment_data, 3.25)

# Summary table using stargazer
stargazer(model_30, model_35,
          type = "text",
          column.labels = c("3.0 vs 2.5", "3.5 vs 3.0"),
          dep.var.labels = "Total Enrollments",
          covariate.labels = c("Treatment (Higher Star Rating)", "Running Va
          title = "RD Estimates using MA 2010 Data",
          omit.stat = c("f", "ser"))
```

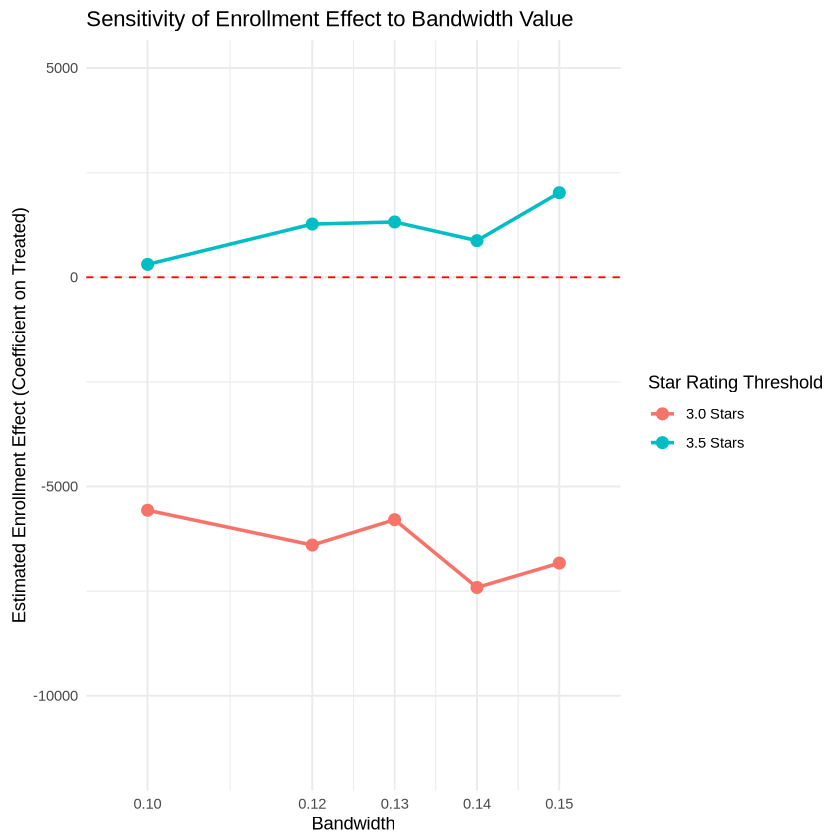
```
RD Estimates using MA 2010 Data
==========================================================
                                   Dependent variable:
                               ---------------------------
                                    Total Enrollments
                                 3.0 vs 2.5    3.5 vs 3.0
                                    (1)            (2)
----------------------------------------------------------
Treatment (Higher Star Rating) -5,792.156***    1,321.590
                                (2,114.001)    (1,649.416)

Running Variable (Distance)     29,235.720**   -6,487.592
                                (14,756.730)  (11,185.220)

Constant                        7,661.156***  3,060.992***
                                (1,535.065)    (1,000.816)

----------------------------------------------------------
Observations                        264            266
R2                                 0.029          0.003
Adjusted R2                        0.021         -0.005
==========================================================
Note:                          *p<0.1; **p<0.05; ***p<0.01
```

Question 7

In [35]:

```
# Define parameters and bandwidths for analysis
thresholds <- c("3.0 Stars" = 2.75, "3.5 Stars" = 3.25)
bandwidths <- c(0.1, 0.12, 0.13, 0.14, 0.15)

# Create a function to run model and extract estimate
run_sensitivity <- function(thresh_val, thresh_label, bw) {
  model_data <- rd_enrollment_data %>%
    mutate(dist = raw_score - thresh_val) %>%
    filter(abs(dist) <= bw) %>%
    mutate(treated = ifelse(dist >= 0, 1, 0))

  fit <- lm(total_enrollment ~ treated + dist, data = model_data)

  # Data frame with estimate and standard error
  broom::tidy(fit, conf.int = TRUE) %>%
    filter(term == "treated") %>%
    mutate(bandwidth = bw, threshold = thresh_label)
}

# Run the model across all combos
sensitivity_results <- map2_dfr(
  rep(thresholds, each = length(bandwidths)),
  rep(names(thresholds), each = length(bandwidths)),
  ~run_sensitivity(.x, .y, bandwidths[match(.y, names(thresholds)) + (match(
)

# Correction for the map: cleaner version
sensitivity_results <- expand.grid(thresh_name = names(thresholds), bw = bar
```

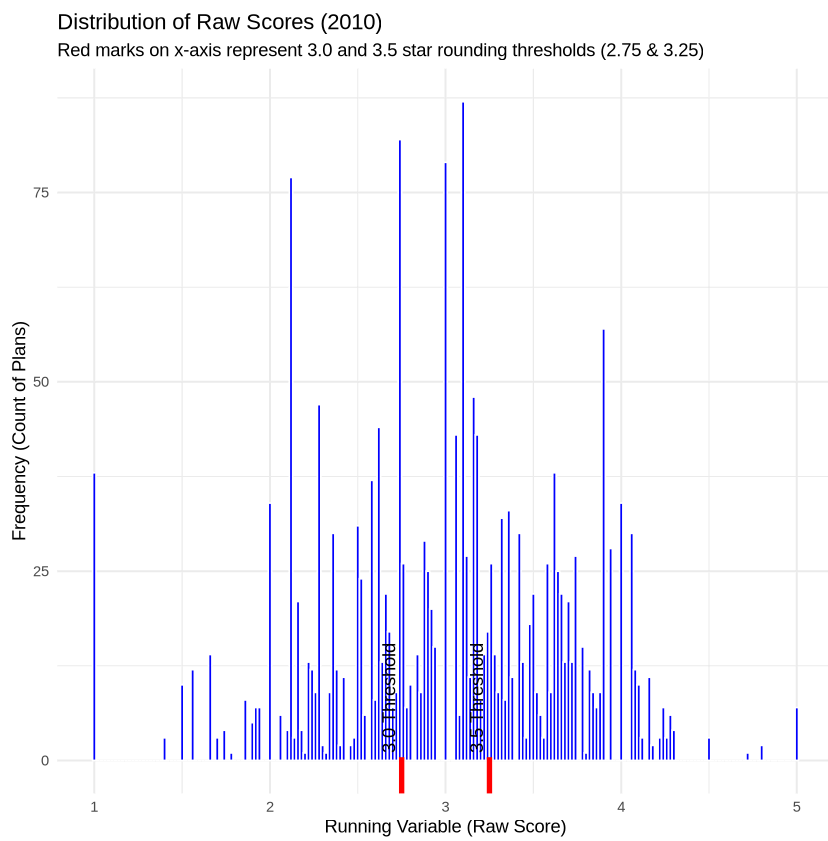
```
  rowwise() %>%
  do(run_sensitivity(thresholds[[.$thresh_name]], .$thresh_name, .$bw))


# Create graph
ggplot(sensitivity_results, aes(x = bandwidth, y = estimate, color = threshc
    geom_line(size = 1) +
    geom_point(size = 3) +
    geom_errorbar(aes(ymin = conf.low, ymax = conf.high), linewidth = 0.005)
    scale_x_continuous(breaks = bandwidths) +
  theme_minimal() +
  labs(
    title = "Sensitivity of Enrollment Effect to Bandwidth Value",
    x = "Bandwidth",
    y = "Estimated Enrollment Effect (Coefficient on Treated)",
    color = "Star Rating Threshold"
  ) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red")
```



As we move from left to right (bandwidth of 0.1 to 0.15), we are including more plans that are further away from the thresholds. At the same time, we are looking at the effect of a 3-star versus a 2.5 star rating (red line) on enrollments and a 3.5 vs 3-star rating (blue line). What do the fluctuations in each of the lines show?

Question 8

In [33]:
```
ggplot(rd_enrollment_data, aes(x = raw_score)) +
geom_histogram(binwidth = 0.02, fill = "blue", color = "white") +

# Mark thresholds
  geom_rug(data = data.frame(raw_score = c(2.75, 3.25)),
           sides = "b",
           color = "red",
           linewidth = 1.5,
           length = unit(0.05, "npc")) + # Adjust length to make the "mark"

  theme_minimal() +
  labs(
    title = "Distribution of Raw Scores (2010)",
    subtitle = "Red marks on x-axis represent 3.0 and 3.5 star rounding thre
    x = "Running Variable (Raw Score)",
    y = "Frequency (Count of Plans)"
  ) +
  annotate("text", x = 2.75, y = 1, label = "3.0 Threshold", color = "black"
  annotate("text", x = 3.25, y = 1, label = "3.5 Threshold", color = "black"
```



The running variable (raw score) before and after the thresholds (red lines) is not notably bunched, showing that there doesn't seem to be much of a difference.

Question 9

In [23]:
```
balance_data <- ma_2010 %>%
  mutate(
```

```
    raw_score = rowMeans(select(., any_of(rating.vars.2015)) %>% select(wher
    is_hmo = ifelse(grepl("HMO", plan_type, ignore.case = TRUE), 1, 0),
    has_partd = ifelse(partd == "Yes", 1, 0)
  ) %>%
  filter(!is.na(raw_score))

# RD Balance Test at 3.0 stars (threshold 2.75)
hmo_3 <- rdrobust(y = balance_data$is_hmo, x = balance_data$raw_score, c = 2
partd_3 <- rdrobust(y = balance_data$has_partd, x = balance_data$raw_score,

# RD Balance Test at 3.5 stars (threshold 3.25)
hmo_35 <- rdrobust(y = balance_data$is_hmo, x = balance_data$raw_score, c =
partd_35 <- rdrobust(y = balance_data$has_partd, x = balance_data$raw_score,

# View results
summary(hmo_3)
summary(partd_3)
```

```
Sharp RD estimates using local polynomial regression.

Number of Obs.                 95203
BW type                       Manual
Kernel                    Triangular
VCE method                        NN

Number of Obs.                 55252        39951
Eff. Number of Obs.            12918         5017
Order est. (p)                     1            1
Order bias  (q)                    2            2
BW est. (h)                    0.125        0.125
BW bias (b)                    0.125        0.125
rho (h/b)                      1.000        1.000
Unique Obs.                    55252        39951

=============================================================================
=
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
=
  Conventional    -0.563     0.007   -84.146     0.000    [-0.576 , -0.550]
        Robust         -         -  -119.253     0.000    [-1.007 , -0.974]
=============================================================================
=
Sharp RD estimates using local polynomial regression.

Number of Obs.                 95203
BW type                       Manual
Kernel                    Triangular
VCE method                        NN

Number of Obs.                 55252        39951
Eff. Number of Obs.            12918         5017
Order est. (p)                     1            1
Order bias  (q)                    2            2
BW est. (h)                    0.125        0.125
BW bias (b)                    0.125        0.125
rho (h/b)                      1.000        1.000
Unique Obs.                    55252        39951

=============================================================================
=
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
=
  Conventional    -0.438     0.021   -20.832     0.000    [-0.479 , -0.396]
        Robust         -         -   -23.131     0.000    [-0.720 , -0.607]
=============================================================================
=
```

In [28]:

```
# Summary table of p-values for 3.0 and 3.5 thresholds
results_table <- data.frame(
  Characteristic = c("HMO Status", "Part D Status"),
  `P-Value at 3.0 Stars (2.75)` = c(hmo_30$pv[1], partd_30$pv[1]),
  `P-Value at 3.5 Stars (3.25)` = c(hmo_35$pv[1], partd_35$pv[1]),
```

```
  check.names = FALSE
)

results_table
```

A data.frame: 2 × 3

Characteristic	P-Value at 3.0 Stars (2.75)	P-Value at 3.5 Stars (3.25)
<chr>	<dbl>	<dbl>
HMO Status	0.000000e+00	1.255537e-19
Part D Status	2.236214e-96	1.728727e-06

Question 10

Looking at the results of questions 5-9, we do not see an effect of star rating on MA enrollments.